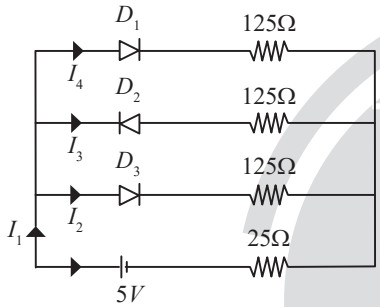
Semiconductor Electronics: Materials, Devices and Simple Circuits

Single correct Type Questions

1. If each diode has a forward bias resistance of 25Ω in the below circuit,



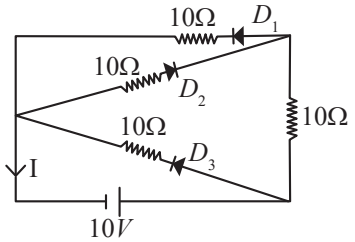
Which of the following options is correct:

[10 Apr, 2023 (Shift-II)]

(*a*) $\frac{I_3}{I_4} = 1$ (*b*) $\frac{I_2}{I_3} = 1$

(*c*) $\frac{I_1}{I_2} = 1$ (*d*) $\frac{I_1}{I_2} = 2$

2. In the given circuit, the current (I) through the battery will be: **[15 Apr, 2023 (Shift-I)]**



(*a*) 1.5 *A* (*b*) 1 *A*

(*c*) 2.5 *A* (*d*) 2 *A*

3. **Statement-I:** When a *Si* sample is doped with Boron, it becomes *P* type and when doped by Arsenic it becomes *N*-type semi conductor such that *P*-type has excess of holes and *N*-type has excess of electrons.

Statement-II: When such *P*-type and *N*-type semi-conductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In light of the above statements, choose the most appropriate answer from the options given below.

[25 Jan, 2023 (Shift-II)]

(*a*) Both Statement-I and statement-II are incorrect

(*b*) Statement-I is incorrect but statement-II is correct

(*c*) Both Statement-I and statement-II are correct

(*d*) Statement-I is correct but statement-II is incorrect

4. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): Photodiodes are preferably operated in reverse bias condition for light intensity measurement.

Reason (R): The current in the forward bias is more than the current in the reverse bias for a *p – n* junction diode.

In the light of the above statement, choose the correct answer from the options given below:

[24 Jan, 2023 (Shift-I)]

(*a*) *A* is false but *R* is true

(*b*) Both *A* and *R* are true but *R* is NOT the correct explanation of *A*

(*c*) *A* is true but *R* is false

(*d*) Both *A* and *R* are true and *R* is the correct explanation of *A*

5. Given below are two statements : one is labeled as Assertion (A) and the other is labeled as Reason (R)

Assertion (A): Photodiodes are used in forward bias usually for measuring the light intensity.

Reason (R): For a p-n junction diode, at applied voltage *V* the current in the forward bias is more than the current in the reverse bias for $|V_z| > \pm V \geq |V_0|$ where V_0 is the threshold voltage and V_z is the breakdown voltage.

Choose the correct answer from the options given below: **[25 Jan, 2023 (Shift-I)]**

(*a*) Both *A* and *R* are true and *R* is the correct explanation of *A*

(*b*) Both *A* and *R* are true but *R* is NOT the correct explanation of *A*

(*c*) *A* is false but *R* is true

(*d*) *A* is true but *R* is false

6. Which of the following Statement-Is not correct in the case of light emitting diodes?

A. It is a heavily doped p-n junction.

B. It emits light only when it is forward biased.

C. It emits light only when it is reverse biased.

D. The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used.

Choose the correct answer from the options given below: **[29 Jan, 2023 (Shift-I)]**

(*a*) *C* and *D* (*b*) *A*

(*c*) *C* (*d*) *B*

7. For using a multimeter to identify diode from electrical components. Choose the correct statement out of the following about the diode: **[28 June, 2022 (Shift-I)]**

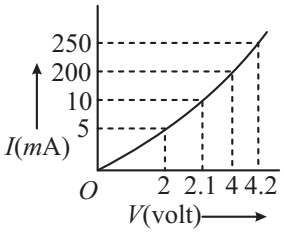
(*a*) It is two terminal device which conducts current in both directions

(*b*) It is two terminal device which conducts current in one direction only

(*c*) It does not conduct current gives an initial deflection which decays to zero.

(*d*) It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

8. The *I–V* characteristics of *p–n* junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltage of 2 V and 4 V respectively, is: **[26 June, 2022 (Shift-I)]**



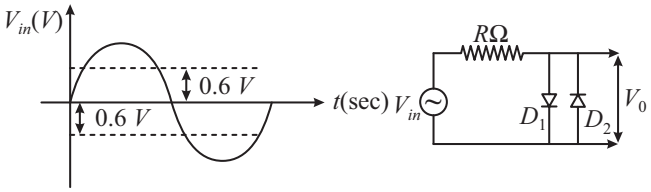
(*a*) 1 : 2

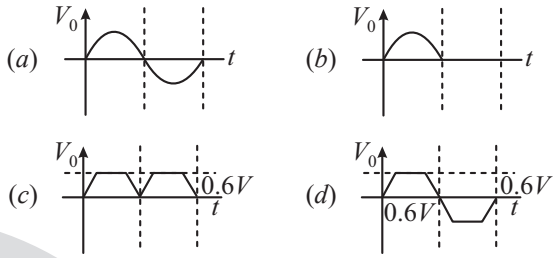
(*b*) 5 : 1

(*c*) 1 : 40

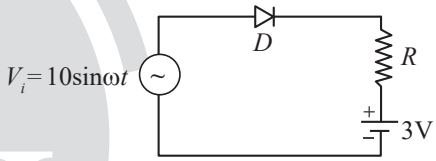
(*d*) 20 : 1

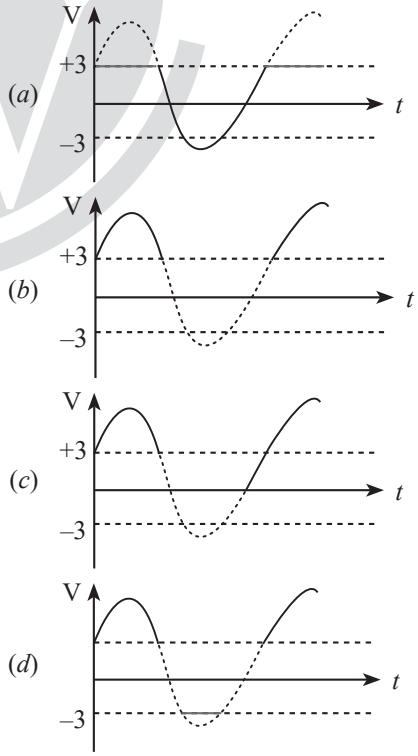
9. In the given circuit the input voltage V_{in} is shown in figure. The cut in voltage of p-n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct? **[28 June, 2022 (Shift-II)]**





10. Choose the correct waveform that can represent the voltage across *R* of the following circuit, assuming the diode is ideal one: **[31 Aug, 2021 (Shift-I)]**





11. For extrinsic semiconductors; when doping level is increased; **[25 Feb, 2021 (Shift-II)]**

(*a*) Fermi-level of p and n-type semiconductors will not be affected.

(*b*) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward.

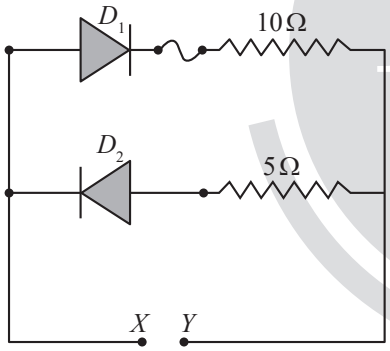
(*c*) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.

(*d*) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F K$ and downward for $T < T_F K$, where T_F is Fermi temperature.

12. Consider a situation in which reverse biased current of a particular $P - N$ junction increase when it is exposed to a light of wavelength ≤ 621 nm. During this process, enhancement in carrier concentration takes place due to generation of hole – electron pairs. The value of band gap is nearly. **[22 July, 2021 (Shift-II)]**

(*a*) 1 eV (*b*) 4 eV (*c*) 0.5 eV (*d*) 2 eV

13. A 5 V battery is connected across the points X and Y. Assume D, and D_2 to be normal silicon diodes. Find the current supplied by the battery if the +*ve* terminal of the battery is connected to point X. **[25 Feb, 2021 (Shift-I)]**



(*a*) ~0.5 A (*b*) ~0.43 A (*c*) ~0.86 A (*d*) 1.5 A

14. LED is constructed from Ga – As - P semiconducting material. The energy gap of this LED is 1.9 eV. Calculate the wavelength of light emitted and its colour.
[$h = 6.63 \times 10^{-34}$ Js and $c = 3 \times 10^8$ ms^{-1}]
[26 Feb, 2021 (Shift-I)]

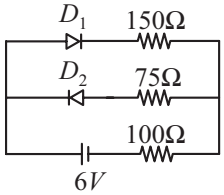
(*a*) 654 nm and orange colour

(*b*) 1046 nm and blue colour

(*c*) 1046 nm and red colour

(*d*) 654 nm and red colour

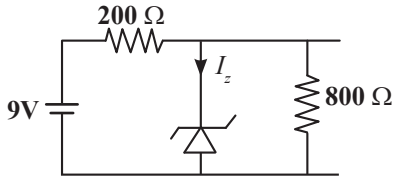
15. The circuit shown below cotains two ideal diodes, each with a forward resistance of 50Ω. If the battery voltage is 6V, the current through the 100Ω resistance (in Amperes) is **[11 Jan, 2019 (Shift-II)]**



(*a*) 0.036 (*b*) 0.020

(*c*) 0.027 (*d*) 0.030

16. The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit. **[8 Apr, 2019 (Shift-I)]**

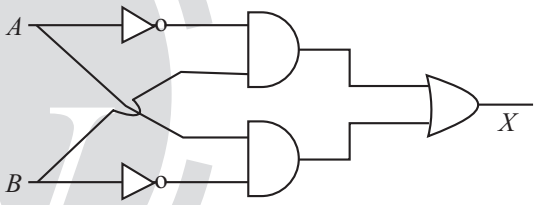


The current I_z, through the Zener is:

(*a*) 10 mA (*b*) 15 mA

(*c*) 7 mA (*d*) 17 mA

17. For the given logic gates combination, the correct truth table will be **[29 Jan, 2023 (Shift-II)]**



(*a*)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(*b*)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

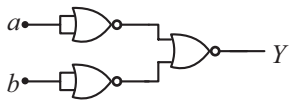
(*c*)

A	B	X
0	0	1
0	1	0
1	0	1
1	1	0

(*d*)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

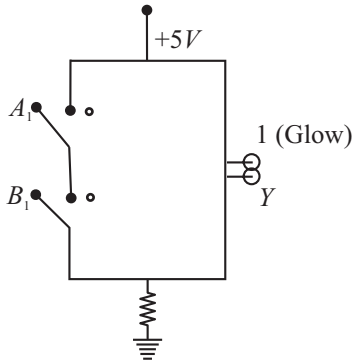
18. The logic performed by the circuit shown in figure is equivalent to: **[11 Apr, 2023 (Shift-I)]**



(*a*) AND (*b*) NAND

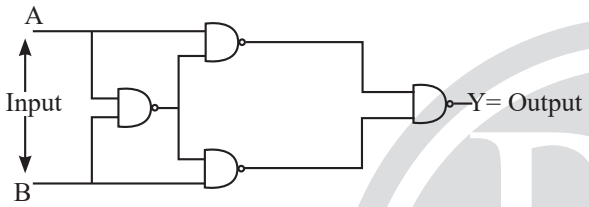
(*c*) OR (*d*) NOR

19. The logic gate equivalent to the given circuit diagram is: **[24 Jan, 2023 (Shift-II)]**



(*a*) OR (*b*) NAND (*c*) NOR (*d*) AND

20. The output *Y* for the inputs *A* and *B* of circuit is given by



Truth table of the shown circuit is: **[30 Jan, 2023 (Shift-II)]**

(*a*)

A	*B*	*Y*
0	0	1
0	1	1
1	0	1
1	1	0

(*b*)

A	*B*	*Y*
0	0	1
0	1	0
1	0	0
1	1	1

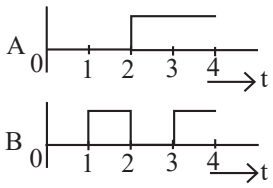
(*c*)

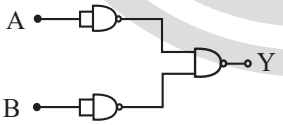
A	*B*	*Y*
0	0	0
0	1	1
1	0	1
1	1	1

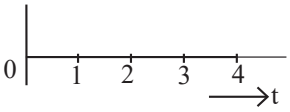
(*d*)

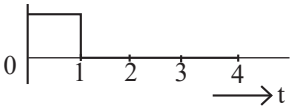
A	*B*	*Y*
0	0	0
0	1	1
1	0	1
1	1	0

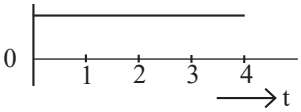
21. For the logic circuit shown, the output waveform at *Y* is: **[08 Apr, 2023 (Shift-I)]**

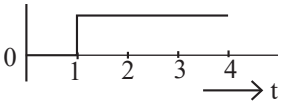




(*a*)


(*b*)


(*c*)


(*d*)


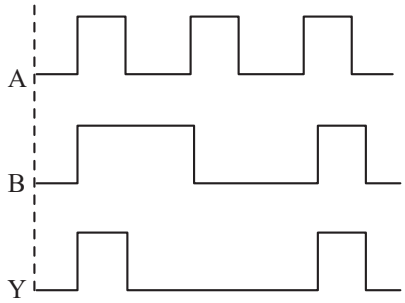
22. A logic gate circuit has two inputs *A* and *B* and output *Y*. The voltage waveforms of *A*, *B* and *Y* are shown below. The logic gate circuit is: **[27 July, 2022 (Shift-I)]**

(*a*) AND gate

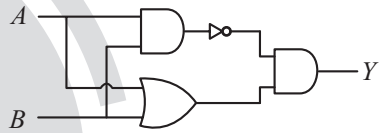
(*b*) OR gate

(*c*) NOR gate

(*d*) NAND gate



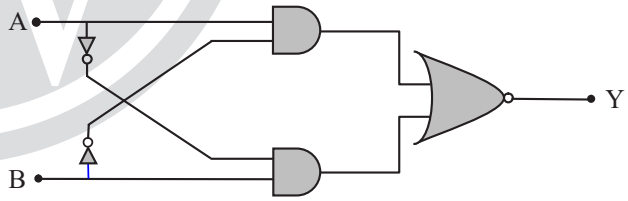
23. Which one of the following will be the output of the given circuit? **[17 Mar, 2021 (Shift-II)]**



(*a*) XOR Gate (*b*) NOR Gate

(*c*) NAND Gate (*d*) AND Gate

24. The truth table for the following, logic circuit is: **[25 Feb, 2021 (Shift-II)]**



(*a*)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(*b*)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(*c*)

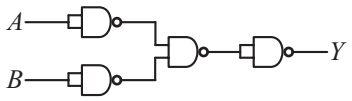
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(*d*)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

25. The following logic gate is equivalent to:

[16 Mar, 2021 (Shift-II)]

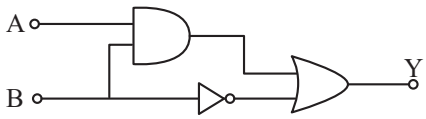


(*a*) NOR Gate (*b*) NAND Gate

(*c*) OR Gate (*d*) AND Gate

26. Find the truth table for the function Y of A and B represented in the following figure.

[27 July, 2021 (Shift-II)]



(*a*)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(*b*)

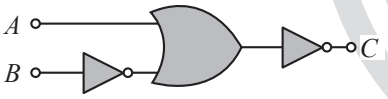
A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(*c*)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

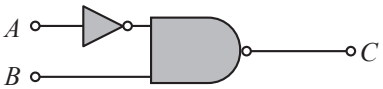
(*d*)

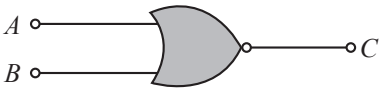
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

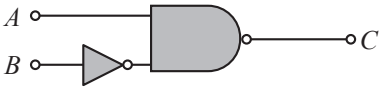
27. 

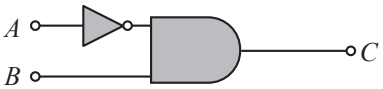
The logic circuit shown above is equivalent to:

[24 Feb, 2021 (Shift-II)]

(*a*) 

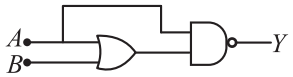
(*b*) 

(*c*) 

(*d*) 

28. The truth table for the circuit given in the figure is

[12 Apr, 2019 (Shift-I)]



(*a*)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(*b*)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(*c*)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

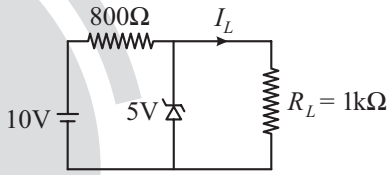
(*d*)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

Integer Type Questions

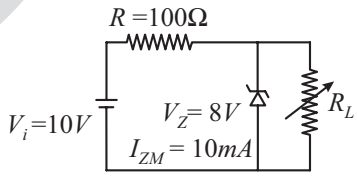
29. In the given circuit, the value of current I_L will be _____ mA. (When $R_L = 1k\Omega$)

[24 June, 2022 (Shift-II)]



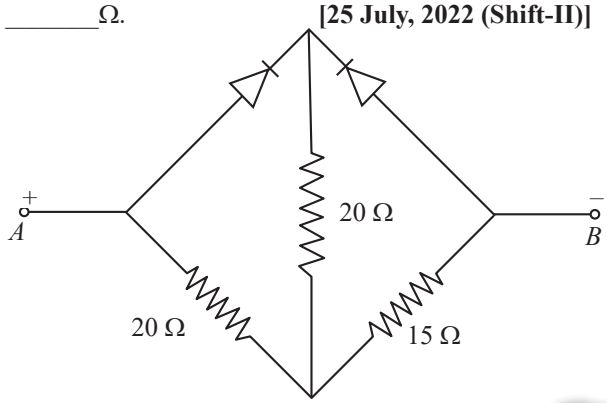
30. A zener of breakdown voltage $V_Z = 8$ V and maximum zener current $I_{ZM} = 10$ mA is subjected to an input voltage $V_i = 10$ V with series resistance R = 100 Ω. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is _____

[28 June, 2022 (Shift-II)]

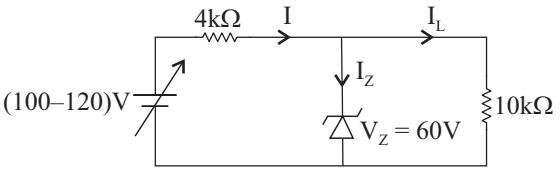


31. Two ideal diodes are connected in the network as shown is figure. The equivalent resistance between A and B is _____Ω.

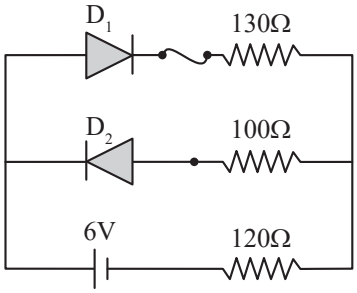
[25 July, 2022 (Shift-II)]



32. In the circuit shown below, maximum zener diode current will be _____mA. **[26 July, 2022 (Shift-I)]**



33. The circuit contains two diodes each with a forward resistance of 50 Ω and with infinite reverse resistance. If the battery voltage is 6 *V*, the current through the 120Ω resistance is _____ m A. **[26 Feb, 2021 (Shift-I)]**

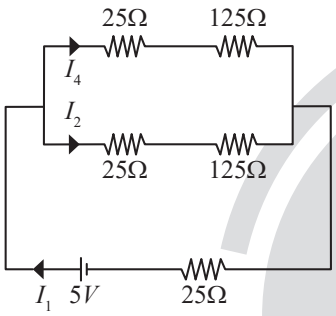


ANSWER KEY

1. (*d*)	**2.** (*a*)	**3.** (*d*)	**4.** (*b*)	**5.** (*c*)	**6.** (*c*)	**7.** (*b*)	**8.** (*b*)	**9.** (*d*)	**10.** (*b*)
11. (*c*)	**12.** (*d*)	**13.** (*b*)	**14.** (*d*)	**15.** (*b*)	**16.** (*a*)	**17.** (*b*)	**18.** (*a*)	**19.** (*b*)	**20.** (*d*)
21. (*d*)	**22.** (*a*)	**23.** (*a*)	**24.** (*a*)	**25.** (*a*)	**26.** (*b*)	**27.** (*d*)	**28.** (*a*)	**29.** [5]	**30.** [2]
31. [25]	**32.** [9]	**33.** [20]							

EXPLANATIONS

1. (*d*) As D_2 is reverse biased. So reduced circuit



$$R_{Total} = \frac{150 \times 150}{300} + 25 = 100\Omega$$

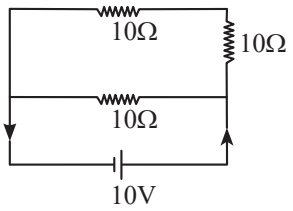
$$I_1 = \frac{5}{100} = 0.05\,A$$

Using KCL $I_1 = I_2 + I_4$

$$I_2 = I_4 = \frac{0.05}{2} = 0.025\,A$$

$$\frac{I_1}{I_2} = \frac{0.05}{0.025} = 2$$

2. (*a*) In the circuit diode, D_1 and D_2 are forward biased and D_2 is reverse biased therefore D_2 Would not allow to the current pass through it.



$$R_{eq} = \frac{(10+10)\times 10}{(10+10)+10} = \frac{20}{3}\Omega$$

$$I = \frac{V}{R_{eq}}$$

$$\therefore I = \frac{10}{20/3} = \frac{3}{2}A = 1.5\,A$$

3. (*d*) When a *P-N* junction diode is formed an electric field is developed from *N*-side to *P*-side due to which barrier potential is created & majority charge carrier can not flow through the junction due to barrier potential so current is zero unless we apply forward bias voltage.

4. (*b*) Photodiodes are operated in reverse biasing, as fractional change in current due to light is more easily measurable in reverse biasing.

5. (*c*) Theory based

A Photodiode is operated in reverse bias condition.

For a P-N junction diode current in forward bias (for $V \geq V_0$) is always greater than the current in reverse bias (for $V \leq V_z$)

Hence, Assertion is false but Reason is true

6. (*c*) LED works when it is forward biased.

7. (*b*) In forward bias diode conducts

In reverse bias it does not conducts.

8. (*b*) $R = \frac{\Delta V}{\Delta i}$

$$\frac{R_1}{R_2} = \frac{\Delta v_1}{\Delta v_2}\frac{\Delta i_2}{\Delta i_1} = \frac{0.1}{0.2} \times \frac{50}{2} = 5$$

9. (*d*) In + ve half cycle

$D_1 \rightarrow F.B.$; $D_2 \rightarrow R.B.$

0 – 0.6 V

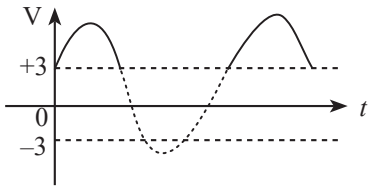
V_{out} same as V_{in}

In –ve half cycle

$D_2 \rightarrow F.B.; D_1 \rightarrow R.B.$

10. (*b*) When $V_i > 3$ volt, $V_R > 0$, because diode will be in forward biased state.

When $V_i \leq 3$ volt, $V_R = 0$, because diode will be in reverse biased state.



11. (*c*) Fermi level of p-type semiconductors will go downward.

12. (*d*) Minimum required energy, $E_{\min} = \dfrac{hc}{\lambda_{\max}}$

$$E_{\min} = \frac{1242 eV - nm}{621 nm} = 2eV$$

13. (*b*) D_1 is in forward bias and D_2 is in reverse bias.

Voltage drop across silicon diode, $V_s = 0.7$

Current supplied by the battery, $i = \dfrac{5-0.7}{10} = 0.43$ A

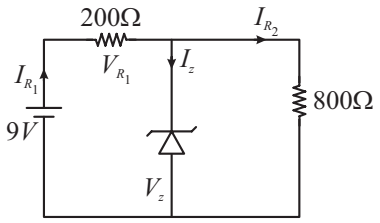
14. (*d*) Wavelength of emitted light,

$$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{1.9 \times 1.6 \times 10^{-19}} = 654\, nm \text{ (Red)}$$

15. (*b*) $I = \dfrac{6}{300} = 0.02A$ (D_2 is in reverse bias)

16. (*a*) $V_{R_1} = 9 - 5.6 = 3.4$ V

$$\therefore\ I_{R_1} = \frac{V_{R_1}}{R} = \frac{3.4}{200} = 17 \text{ mA}$$

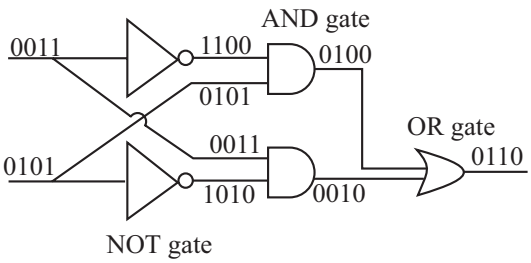


$I_z = I_{R_1} - I_{R_2}$

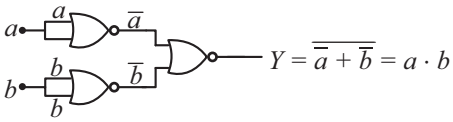
$$I_z = \left(17 - \frac{5.6}{800}\right) \text{mA}$$

$I_z = (17 - 7)$ mA $= 10$ mA

17. (*b*)



18. (*a*)



$Y = \overline{\bar{a} + \bar{b}} = a \cdot b$ (using Demorgon theorem)

This will be AND gate.

19. (*b*)

A_1	B_1	Y
0	0	1
0	1	1
1	0	1
1	1	0

As the output $Y = \overline{A_1 \odot B_1}$

$\Rightarrow$ *NAND* gate

20. (*d*) Given circuit act like a XOR.

21. (*d*) $Y = \overline{\bar{A} \cdot \bar{B}} = \bar{\bar{A}} + \bar{\bar{B}} = A + B$

So truth table,

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

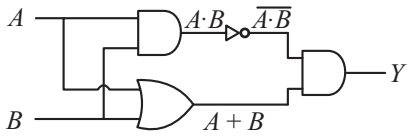
According to value of output Y, option d will be correct output waveform.

22. (*a*)

A	B	Y
0	0	0
1	1	1
0	1	0
1	0	0

According to the truth table the logic is AND gate.

23. (*a*) The result of a XOR gate is high (1) just when precisely one of its bits of inputs is high (1). In the event that both of a XOR gate's bits of inputs are low (0), or then again on the off chance that both of its bits of inputs are high (1), the result of the XOR gate is low.



$y = (\overline{A\cdot B})\cdot(A+B)$

$= (\overline{A}+\overline{B})\cdot(A+B)$

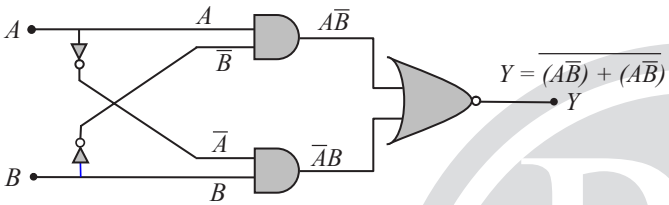
$= \overline{A}A + \overline{A}B + A\overline{B} + \overline{B}B$

$= 0 + \overline{A}B + A\overline{B} + 0$

$y = \overline{A}B + A\overline{B}$

Which is X O R gate

24. (*a*)

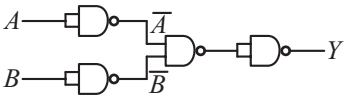


$Y = \overline{(A\overline{B})+(\overline{A}B)} = \overline{\left((\overline{A}B)\right)}\,\overline{\left((\overline{A}B)\right)}$

$Y = (\overline{A}+\overline{\overline{B}})+(\overline{\overline{A}}+\overline{B}) = \left((\overline{A}+B)\right)\left((A+\overline{B})\right)$

$= AB + \overline{A}\overline{B}$

25. (*a*) The NOR gate is a computerized rationale door that carries out sensible NOR- it acts as per reality table to one side. A HIGH result (1) results if both the contributions to the entryway are LOW (0); in the event that one or both information is HIGH (1), a LOW result (0) results. NOR is the consequence of the nullification of the OR administrator.



Output of first NAND gate with input

$= \overline{A\cdot A} = \overline{A}+\overline{A} = \overline{A}$

Similarly, Output of first NAND gate with input B

$= \overline{B}$

$Y_1 = \overline{\overline{A}\cdot\overline{B}} = \overline{\overline{A}}+\overline{\overline{B}} = A+B$

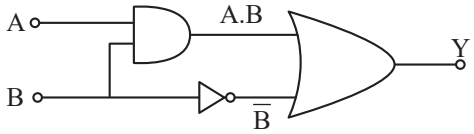
$Y = \overline{(A+B)\cdot(A+B)} = \overline{(A+B)}+\overline{(A+B)} = \overline{(A+B)}$

So, given logic gates circuit is a NOR gate.

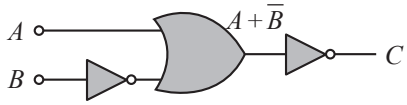
26. (*b*) Here $Y = A\times B+\overline{B}$

So the output will be as given in the below table

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

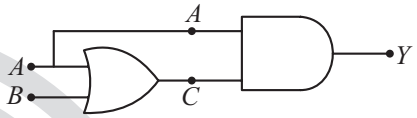


27. (*d*)



$C = \overline{A+\overline{B}} = \overline{A}.\overline{\overline{B}} = \overline{A}\,B$

28. (*a*)

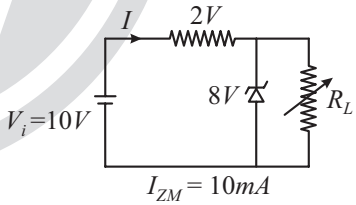


C = A + B and Y = $\overline{A.C}$

A	B	C = (A + B)	A.C.	Y = $\overline{A.C}$
0	0	0	0	1
0	1	1	0	1
1	0	1	1	0
1	1	1	1	0

29. [5] $I_L = \dfrac{V}{R_L} = \dfrac{5}{1000} = 5mA$

30. [2]



$I = \dfrac{2}{100} = 20\,mA$

$V_L = I_L R_L$

$8 = 10\times10^{-3}\times R_{L_{max}}$

$\dfrac{4}{5}\times10^3 = R_{L_{max}}$

$I = I_Z + I_L$

$I_L = 10\ mA$

If $I_Z = 0$

$\because\ I_{L_{max}} = 20mA$

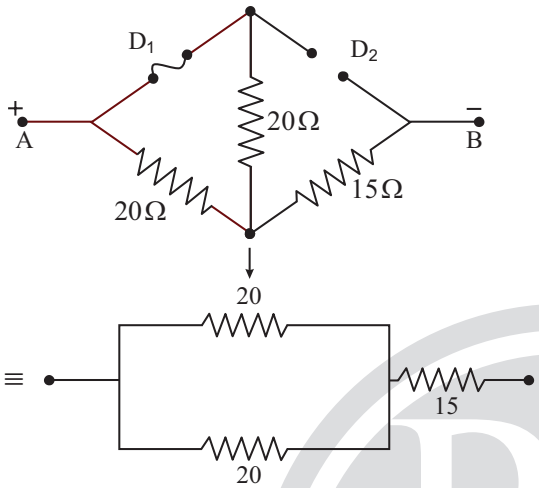
$V_L = I_{L_{max}}\times R_{L_{max}}$

$$\frac{8}{20}\times 10^3 = R_{L_{\min}}$$

$$400 = R_{L_{\max}}$$

$$\frac{R_{L_{\max}}}{R_{L_{\min}}} = \frac{800}{400} = 2$$

31. [25]



Here diode-D_1 is forward biased and diode-D_2 is reversed biased, Equivalent resistance between A & B.

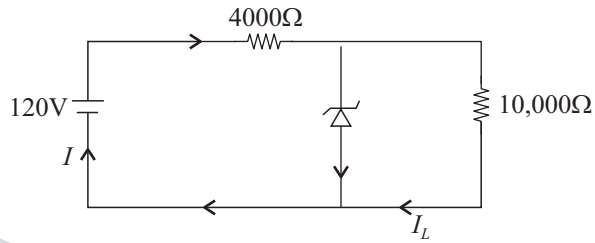
$$R_{eq} = \frac{20\times 20}{20+20} + 15 = 25\ \Omega$$

32. [9] Current in the load, $I_L = \frac{60}{10,000} = 0.006A$

$$I = \frac{120-60}{4000} = 0.015A$$

Thus, maximum zener current, $I_Z = I - I_L$

$= 0.015 - 0.006 = 0.009 = 9\text{mA}$



33. [20] D_1 is forward biased & D_2 is reverse biased.

The current through the 120Ω resistance

$$I = \frac{6}{120+130+50} = \frac{6}{300} = 2\times 10^{-2}\ A = 20\times 10^{-3}\ A$$

$$= 20\,mA$$